

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

5th Semester of B. Pharm. Examination

University Theory Examination December 2015

PH306 Pharmaceutical Biotechnology

Date: 04/12/2015, Friday

Time: 10:00a.m to 1:00p.m

Maximum Marks: 80

Instructions:

1. Figures to right indicate marks.
2. Section wise separate answer book to be used.
3. Draw neat sketches wherever necessary.

SECTION- I

Q 1 Attempt any TWO of the following

- A i) Enlist types of mutation . How mutation in bacteria get repaired? 05
- ii) Explain Particle Gun delivery for gene transfer with diagram 05
- B Write a note on DNA replication with suitable diagram 10
- C What is recombinant DNA Technology? Write various steps to make recombinant DNA in brief. Explain cloning of cDNA 10

Q 2 Attempt any TWO of the following

- A i) Enlist enzymes and vectors involved in gene cloning. Discuss Clonning eukaryotic DNA into bacterial plasmid 5
- ii) Write a note on polymerase chain reaction 5
- B Discuss conjugation and protoplast fusion in bacteria. 10
- C Differentiate transformation and transduction process in brief . Give mechanism of transduction in bacteria with diagrammatic representations 10

SECTION- II

Q 3 Attempt any TWO of the following

- A Enlist the methods which are used for production of vinegar. Describe the following methods 10
- i) Orleans process
- ii) Submerged generator process
- B i) Classify various methods for enzyme immobilization. Explain any one method in detail 5
- ii) Write a note on amylase enzyme 5

- C Discuss salient features of Single cell Protein production. Give advantages of SCP 10
- Q 4 Attempt any TWO of the following
- A Describe gluconic acid acid production. Write its application 10
- B i) Draw the pathway of Lactic Acid Production from Glucose Using Lactobacillus Bacteria. Explain any one method for extraction and recovery of it 5
- ii) Draw the biosynthetic pathway of itaconic acid 5
- C Explain various biological precautions need to consider for safety in laboratory 10
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